

Education

University of Connecticut <i>PhD Candidate in Statistics, Department of Statistics</i> Courses: Design of Experiments, Statistical Inference, Linear Models	Sep 2024 – Present <i>Storrs, CT, U.S.</i>
Renmin University of China <i>Master of Science in Applied Statistics, Institute of Statistics and Big Data</i> Courses: Clinical Trial Statistical Methodology, Causal Inference, Survival Analysis	Sep 2022 – Jun 2024 <i>Beijing, China</i>
Xi'an Jiaotong University <i>Bachelor of Economics in Finance, School of Economics and Finance</i> Courses: Probability Theory, Mathematical Statistics, Econometrics	Sep 2018 – Jun 2022 <i>Xi'an, China</i>

Experience

Astellas Pharma <i>Biostatistician Intern</i> Topic: Closed-Form Evaluation of Correlated PFS and OS Group Sequential Designs - Developed an event-driven group sequential design for correlated progression-free survival (PFS) and overall survival (OS) under the Fleischer model, supporting efficacy monitoring, futility stopping, and piecewise-uniform accrual. - Derived a closed-form joint correlation matrix for sequential PFS and OS log-rank statistics, established a multivariate-normal asymptotic approximation, and calculated boundaries in three scales (z-score, p-value, and hazard ratio), marginal power, and gatekeeping joint power. - Built an R package and integrated R Shiny application that automate trial-design specification, closed-form analysis, visualization, and Monte Carlo simulation-based validation, enabling efficient and reproducible design evaluation.	May 2026 – Aug 2026 <i>Remote</i>
Department of Public Health Sciences, UConn Health <i>Research Assistant</i> Topic: Longitudinal Analysis of Depression Outcomes in Large-Scale Medicare Claims Data - Processed 3 billion+ Medicare claim datasets using high-performance big-data tools in R, including <code>fst</code> and <code>data.table</code>, enabling efficient extraction and linkage of depression patient cohorts across multiple years. - Constructed a 2 million+ patient longitudinal cohort of Medicare beneficiaries with depression by integrating claims histories, chronic condition indicators, and demographic covariates. - Applied survival analysis methods to identify key clinical and demographic factors associated with the onset and progression of depression among Medicare patients.	Sep 2025 – Dec 2025 <i>Farmington, CT, U.S.</i>
Center for Population Health, UConn Health <i>Research Assistant</i> Topic: Survival and Machine Learning Analysis of Suicide Risk Using Clinical Data - Analyzed 5 million+ patient records spanning multi-year clinical encounters. Engineered features from demographic covariates, ICD diagnostic codes, chronic conditions, and medication histories. - Compared logistic regression, Weibull and Cox proportional hazards models to quantify risk factors associated with suicide rates. Identified high-risk covariates based on model coefficients and hazard ratios. - Developed ML-based suicide risk prediction pipelines using Random Forest, SVM, and Neural Networks, achieving a 20% improvement in AUC over baseline models and delivered model outputs as risk scores to support clinical decision-making workflows.	Jan 2025 – Jun 2025 <i>Farmington, CT, U.S.</i>
Connecticut Children’s Medical Center <i>Research Assistant</i> Topic: Time-Series Forecasting of Emergency Department Patient Volume - Preprocessed and visualized emergency department census data from Connecticut Children’s Medical Center, using heatmap calendar plots to reveal key periodic patterns and temporal trends in patient flow. - Collected external covariates (temperature, air pollution, water quality) from public APIs, integrated them with census data, and performed data merging for downstream modeling and analysis. - Utilized Fourier series to construct a harmonic dynamic regression model to estimate different periodic cycles, combined with an ARIMAX framework to model non-periodic short-term variations, improving MAE/MAPE/RMSE by 30% over baseline time-series models.	Aug 2024 – Jun 2025 <i>Hartford, CT, U.S.</i>

Technical Skills

Programming: R (dplyr, tidyverse, ggplot2); Python (Numpy, Pandas, PyTorch); Rcpp; SAS; MySQL; Git; Linux
Languages: Chinese(Native); English(Fluent - TOEFL: 100 (R28 L26 S22 W24))